**Tutorial 2. Atoms**

**Exercise 1** ***(App)***

Hydrazine, ammonia and hydrogen azide all contain only nitrogen and hydrogen. The mass of hydrogen that combines with 1.00 g of nitrogen for each compound is $1.44 \times 10^{-1}$ g, $2.16 \times 10^{-1}$ g, and $2.40 \times 10^{-2}$ g, respectively. Show how these data illustrate the law of multiple proportions?

**Exercise 2.** ***(Und)***

How many protons and neutrons are in the nucleus of each of the following atoms? In a neutral atom of each element, how many electrons are present?

**a.** $^{79}Br$ **d.** $^{133}Cs$
**b.** $^{81}Br$ **e.** $^{3}H$
**c.** $^{239}Pu$ **f.** $^{56}Fe$

**Exercise 3.** ***(App)***

Complete the following table:

| Symbol | Number of Protons in Nucleus | Number of Neutrons in Nucleus | Number of Electrons | Net Charge |
|---|---|---|---|---|
| $^{238}_{92}U$ | | | | |
| | 20 | 20 | | 2+ |
| | 23 | 28 | 20 | |
| $^{89}_{39}Y$ | | | | |
| | 35 | 44 | 36 | |
| | 15 | 16 | | 3− |

**Exercise 4.** ***(Know)***

Identify the element represented by each of the following symbols and give the number of protons and neutrons in each: (a) $\left(^{11}_{5}X\right)$ (b) $\left(^{75}_{33}X\right)$ (c) $\left(^{86}_{36}X\right)$ (d) $\left(^{67}_{30}X\right)$

**Exercise 5.** ***(Und)***

The isotope of an unknown element, X, has a mass number of 79. The most stable ion of the isotope has 36 electrons and forms a binary compound with sodium having a formula of $Na_2X$. Which of the following statements is (are) true? For the false statements, correct them.

**(i)**. The binary compound formed between X and fluorine will be a covalent compound.

**(ii)**. The isotope of X contains 38 protons

**(iii)**. The isotope of X contains 41 neutrons

**(iv)**. The identity of X is strontium, Sr.

**Exercise 6. (*App*)**

Three isotopes of silicon occur in nature: $^{28}Si$ (92.23%), atomic mass 27.97693 amu; $^{29}Si$ (4.68%), atomic mass 28.97649 amu; and $^{30}Si$ (3.09%), atomic mass 29.97377 amu. Calculate the atomic weight of silicon.

**Exercise 7. *(App)***

A 2.144-g sample of phosgene, a compound used as a chemical warfare agent during World War I, contains 0.260 g of carbon, 0.347 g of oxygen, and 1.537 g of chlorine. What is the empirical formula of this substance?

**Exercise 8. *(Adv App)***

Determine the empirical and molecular formulas of each of the following substances:

a. Caffeine, a stimulant found in coffee, contains 49.5% C, 5.15% H, 28.9% N, and 16.5% O by mass and has a molar mass of 194 g/mol.

b. Monosodium glutamate (MSG), a flavor enhancer in certain foods, contains 35.51% C, 4.77% H, 37.85% O, 8.29% N, and 13.60% Na, and has a molar mass of 169 g/mol.